

Project Charter: Capstone Project Management & Repository System (CPMRS App)

This Project Charter formally authorizes the development of the **Capstone Project Management & Repository System (CPMRS App)** for **The Neotia University**.

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1. Project Purpose

The purpose of the **CPMRS App** (`cpmrs-app`) is to establish an enterprise-grade full-stack portal managing the end-to-end lifecycle of B.Tech final-year capstone projects at **The Neotia University**.

2. Business Need

Managing capstone projects across academic departments (**Cyber Security**, **AIML**, **Data Science**, **CSE**, **ECE**) manually caused project topic duplication, uneven faculty guide workloads, lack of progress tracking, and audit delays during accreditation reviews (NBA, NAAC).

3. Project Objectives

- **100% Process Digitization:** Automate student registration, dual login (Email/Roll No), group formation, 5-digit join codes, topic proposals, guide quotas, and document submissions.
- **Uniqueness & Governance:** Eliminate duplicate project titles and enforce maximum group sizes ($N_{max} = 5$) and guide capacity limits (G_{max}).
- **Accreditation Ready:** Provide instant project archiving, department templates, and activity audit trails (`AuditLog`).

4. Project Scope

In-Scope

- **Group Management:** Group creation, 5-digit join code teaming, join request queues, leadership transfer.
- **Proposals & Allocations:** Faculty proposals, student applications/pitches, guide assignment under quota limits.
- **Submissions Engine:** Synopsis upload, interim milestone progress reports, final thesis deliverables (code ZIP, report PDF, presentation slides).
- **Governance:** Next.js Edge Middleware route guards, role isolation (Student, Faculty Guide, Department Coordinator, HOD, Admin, Super Admin), system settings, audit logs, and templates.

5. Major Deliverables

1. **Full-Stack Application (`cpmrs-app`):** Built with Next.js 16 App Router & React 19.
2. **Database & Schema:** 16 Prisma models mapped to PostgreSQL.
3. **API Modules:** 11 App Router API route handler directories in `src/app/api/` .
4. **Documentation:** Architecture specifications, SRS, ER diagrams, and user manuals.

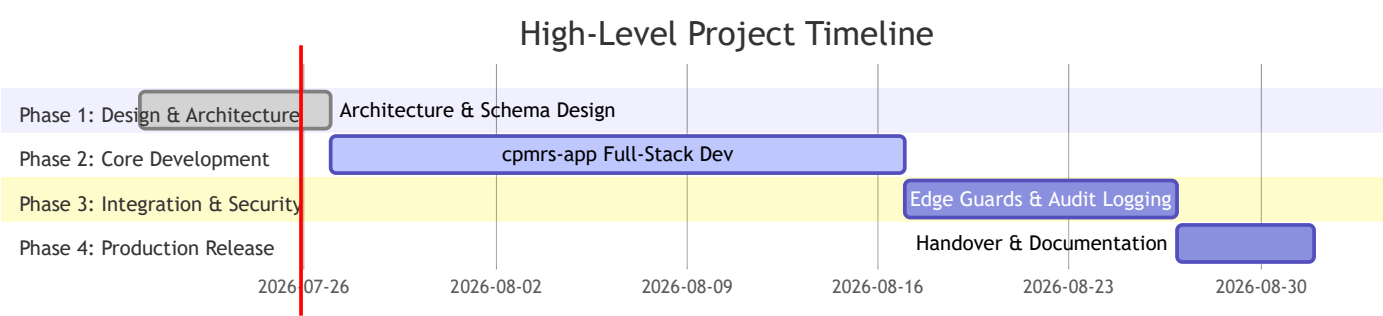
6. Technology Stack

- **Framework:** Next.js 16 (App Router + Turbopack)
- **Frontend:** React 19, Tailwind CSS 4, Lucide Icons
- **Language:** TypeScript 5
- **Database & ORM:** PostgreSQL, Prisma 6 ORM with Accelerate connection pooling
- **Auth:** Edge Middleware (`middleware.ts`), HTTP-only JWT cookies (`jsonwebtoken` , `jose`), `bcryptjs`

7. Stakeholders & Roles

- **University Leadership & HODs:** Department oversight, guide quota reviews, audit exports.
- **Faculty Guides:** Topic proposals, group mentorship, synopsis & milestone evaluations.
- **Students:** Dual login, 5-digit join code teaming, project applications, deliverable uploads.
- **Admin:** Roster management, user accounts, system configuration, audit logs.

8. High-Level Timeline



9. Project Approval

Accepted and approved for implementation at **The Neotia University**.